

Chapter 9

Rod Oscillation

In physics, it is common for physical formulations to be derived through theoretical foundations before being tested by experiments. These experiments typically involve a series of measurements of physical quantities, which are then analyzed to achieve the objectives of the experiment. From this statement, it implies that experimental results are generally considered more reliable than theoretical result.

In this laboratory topic, the usual approach is reversed, namely from the experiment which is aimed at producing a physical formulation. This experiment has two main objectives. First, to determine the relationship between the oscillation period of a rod (T), the distance between the strings (D), and the length of the string (L). Second, to develop skills in constructing and analyzing graphs. The value of this experiment lies demonstrating that physics experiments are not only conducted to measure physical quantities, but can also be used to derive physical's law, based on the understanding of graphs and the theory of uncertainty.

9.1 Objectives

1. To learn how to apply and interpret graph.
2. To determine the relationship between the oscillation period and the length of the string graphically.

9.2 Tools

1. Stand rods (2 pieces), 70 cm support rod (1 piece), and 50 cm oscillating rod (1 piece).
2. Thread as string for rod oscillation, 1 meter (2 strands).
3. Measuring tape (1 piece).
4. Set square ruler (1 piece).
5. Stopwatch (1 piece).

9.3 Theoretical Foundation

The occurrence of the rod oscillation is caused by a restoring force directed toward its equilibrium point. This restoring force causes the oscillation to continue, forming simple harmonic motion (SHM), when air friction is small enough to be neglected. The oscillation has a period (T), string length (L), and distance between the strings (D) (Figure 9.1). The dependence of T on D and L is assumed to be:

$$T = \alpha D^a L^b \quad (9.1)$$

In equation 9.1, α , a , and b are constants that can be determined experimentally.

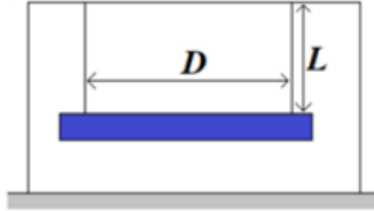


Figure 9.1: Experimental setup of bar oscillation, initiated by a small-angle displacement of the bar

9.4 Experimental Methods

9.4.1 Experimental set-up and equipment

This experimental set-up consists of a horizontally extended rod tied with two parallel strings (spaced at a distance D), with

the upper parts attached to a rod supported by two stands (Figure 9.2(a)). The length of the strings (L) is oscillated, and the period is measured every 5 oscillations, as many as 2 times (Figure 9.2(b)). There are systematic errors in D and L related to the reference points of their measurements (Figure 9.3).

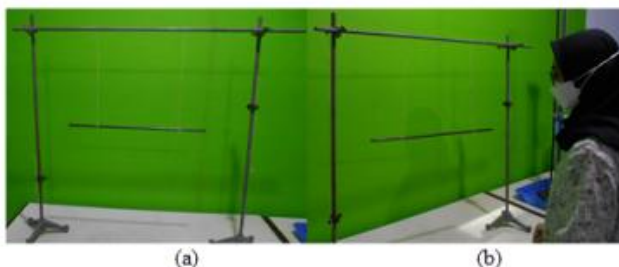


Figure 9.2: (a) Set-up of rod oscillation, (b) When the rod is oscillating



Figure 9.3: Direction of oscillation and suspected occurrence of systematic error

9.4.2 Procedure

For the constant values α , a and b , it is (reasonably) assumed that the relationship between T and D and L takes the form: $T = \alpha D^a L^b$. The objective of this experiment is to determine α , a , and b accurately. The procedure is performed in two separate steps: (1) D is varied while L is kept constant; (2) L is varied while D is kept constant.

Step-1: D is varied while L (60 cm)

1. The set-up is assembled (Figure 9.1), with $D = 40$ cm, and the rod is displaced to oscillate. The oscillation period of the rod is measured every 5 oscillations. This is performed twice so that the period values (at $D = 40$ cm and $L = 60$ cm) is obtained as: $T = \bar{T} \pm \Delta T$
2. Step 1 is repeated with $D = 30, 20$, and 10 cm, respectively.
3. Based on the formulation: $T = (\alpha(60)^b)D^a = KD^a$, or $\ln T = a \ln D + \ln K$. Next, a graph of $\ln T$ vs $\ln D$ is plotted, from which the value of a is obtained (presented without uncertainty, as it is only used as supporting evidence for the exponent of D).

Step-2: L is varied while D (30 cm)

1. The same set-up (Figure 9.1) is used, with $L = 60$ cm, and the rod is displaced to oscillate. The oscillation period of the rod is measured every 5 oscillations. This is done in two measurements to obtain the value of the period (at $D = 30$ cm and $L = 60$ cm): $T = \bar{T} \pm \Delta T$.
2. Step-2 is repeated for $L = 55, 50, 45, 40, 35, 30, 25$ and 20 cm, respectively.
3. Based on the formulation: $T = (\alpha(30)^a)L^b = CL^b$, or $\ln T = b \ln L + \ln C$. Next, a graph of $\ln T$ vs $\ln D$ is plotted, from which the value of a is obtained (presented without uncertainty, as it is only used as supporting evidence for the exponent of L).

Step-3: Determination of α